

predictability barrier (SPB) in May to July, when the forecast skill decreases rapidly (vertical blue lines in Fig. 2o). The SPB is much less pronounced in the XRO model, which maintains a 0.5 correlation skill up to 16 months for all different initial times (Fig. 2p). The superior efficacy of XRO in ENSO forecasting is further illustrated by the root mean square error metric (Supplementary Fig. 10).

Sources outside the tropical Pacific

The XRO formulation allows us to explicitly isolate and quantify the roles of different mode interactions in ENSO's dynamical behaviour and predictability. Three previous approaches have been used to assess the impact of climate variability in various ocean basins on ENSO predictability, using CGCMs, intermediate complexity models and/or conceptual models. They include: (1) partial initialization experiments, which set the ocean initial conditions for a specific basin to the model climatology while using the observed initial conditions everywhere else^{21,28}; (2) partially coupled experiments, which apply strong SST restoring towards the model climatology in a specific region during the model integration, while keeping the atmosphere and ocean fully coupled elsewhere^{22,24,28} and (3) relaxing towards observations experiments, in which model SSTs are strongly nudged towards observations in a specific region, while elsewhere the model is fully coupled^{23,26}. We apply these strategies to our XRO model in corresponding sets of ENSO reforecasting sensitivity experiments: (1) uninitialized experiments (referred to as U_j), (2) decoupled experiments (D_j) and (3) relaxation towards observations experiments (R_j) ('Quantitative reforecasting experiments' section in the Methods and Extended Data Table 1). We further investigate the total contribution of all the modes in each ocean basin to ENSO's predictability by grouping modes together: the extratropical Pacific Ocean (ExPO) includes NPMM and SPMM; the IO IOB, IOD and SIOD; and the AO TNA, ATL3 and SASD. The ExPO + IO + AO experiments demonstrate the combined effects of all the non-ENSO modes.

All the sensitivity experiments qualitatively indicate that coupling information from the ExPO, IO and AO basins enhances ENSO forecast skill (Fig. 3a), consistent with previous findings^{23,24,26,28,35}. However, only the uninitialized experiment framework is a suitable approach to quantify the nearly additive relative contributions of each basin to ENSO forecast skill (Extended Data Fig. 3a,d,e) without artificially overestimating the contribution of climate variability in other basins to ENSO predictability (Extended Data Fig. 3b–e). Therefore, hereafter we use the uninitialized experiment framework to quantify the impact of each individual basin's or mode's initial condition on subsequent ENSO forecast skill.

Allowing for climate mode interactions enhances ENSO forecast skill, and significantly weakens the SPB with an improvement of correlation skill up to 0.2 ($P < 0.08$, Fig. 3b). The enhancement of ENSO forecast skill from climate mode interactions is primarily through the initial condition memory of the different climate modes, demonstrated by the large difference between control and the uninitialized ExPO + IO + AO experiment (Fig. 3c and Supplementary Fig. 11a). The initial states of the other modes can persist for a few months and effectively affect ENSO in specific seasons. By contrast, as evidenced by the minor differences between uninitialized ExPO + IO + AO experiment and decoupled ExPO + IO + AO experiment, the coupled feedbacks with these modes induced by ENSO's initial state only slightly reinforce and accelerate phase transition of ENSO events (Supplementary Fig. 11b). This results in an increase in forecast skill during the ENSO transition phase (Jun⁺¹–Sep⁺¹ targets, Fig. 3d) but a decrease in forecast skill during the ENSO peak phases (Nov⁺¹–Mar⁺¹ targets, Fig. 3d). Further reforecasting experiments ('Losing memory experiments' section in the Methods and Extended Data Fig. 4) confirm that gradually preserving the initial condition memory of climate modes outside the equatorial Pacific incrementally improves ENSO forecast skill from that of the nRO to that of the XRO.

We further illuminate the roles of individual basins in ENSO predictability by comparing the difference between the control and uninitialized experiments for the ExPO, IO and AO basin experiments (Fig. 3e–g). The contributions of each basin have strong seasonality. For instance, the effect of ExPO initialization is most pronounced when forecasts start from November to June, and target December to March when the ENSO signal is large (Fig. 3e). This effect is dominated by the NPMM initialization, whereas the SPMM initialization is less effective (Extended Data Fig. 5a,b). By contrast, the effect of IO initialization is most pronounced when forecasts start from July to November, the time of the year when the IOD develops and peaks (Fig. 3f). The IO effect is dominated by the IOD, with a secondary contribution from the IOB, and the SIOD playing only a minor role (Extended Data Fig. 5c–e). This result is in contrast with the previous finding based on the decoupled linear inverse model experiments¹⁴, which indicated that the IOB plays a more significant role than the IOD in weakening the ENSO SPB. The discrepancy may stem from the lack of seasonality and nonlinearity in their model, along with potential overestimations arising from their decoupled model experiment strategy. The AO also results in a weakening of the ENSO SPB when forecasts are initialized from December to April (Fig. 3g), with main contributions from the TNA and SASD, whereas ATL3 initialization has a negligible effect (Extended Data Fig. 5f–h). These contributions of mode interactions to ENSO forecast skill are further supported by the root mean square error metric (Supplementary Fig. 12).

ENSO intensification from remote sources

Next, we quantify the roles of mode interactions on the individual ENSO event reforecasts, illustrated by the time series of predicted Niño3.4 SSTs for the XRO, decoupled ExPO + IO + AO ($D_{\text{ExPO+IO+AO}}$) and uninitialized ExPO + IO + AO ($U_{\text{ExPO+IO+AO}}$) experiments at lead times of 0–21 months (Fig. 4a–c). The zero lead time refers to the observed values. The Niño3.4 forecasts in the $U_{\text{ExPO+IO+AO}}$ experiment closely resemble those of the $D_{\text{ExPO+IO+AO}}$ experiment, again indicating that the skill improvement in the control XRO arises from the memory of the other climate mode initializations. These two sensitivity reforecasts can predict the El Niño and La Niña event occurrences at lead times of 3–9 months and usually underestimate the amplitude of Niño3.4 SSTs. The XRO systematically outperforms the uninitialized and/or decoupled ExPO + IO + AO experiments with more accurate amplitude prediction of Niño3.4 SSTs and extended skilful prediction of El Niño and La Niña event occurrences at longer lead times of 6–18 months (Fig. 4a). For instance, the 1986–1987 El Niño event could be predicted 18 months in advance with XRO in our hindcast, as opposed to only 6 months in advance with uninitialized or decoupled ExPO + IO + AO experiments.

To better understand the influence of a specific climate mode on individual ENSO events, we examined the differences in ENSO SSTs and WWV anomalies between control and uninitialized experiments for the 1997–1998 El Niño and 1998–2000 triple La Niña episodes (Fig. 4d–k) as well as for the full period (Extended Data Fig. 6). The ENSO forecast differences due to the initialization of other modes are pronounced when those SSTs have sufficiently large amplitudes and during the season in which their interaction with ENSO is relatively strong. These effects of the non-ENSO modes usually last longer than their own SSTA persistence, indicating the activation of ENSO coupled recharge–discharge feedbacks as shown by the ENSO SSTA and WWV anomalies alternating with a few months lag.

In the extratropical Pacific, positive SSTs for both the NPMM and SPMM in boreal spring can enhance ENSO SST warming 6–9 months later (Fig. 4d,h). However, the underlying mechanisms differ for the two different hemispheres. The NPMM warming leads to recharged WWV anomalies and subsequent ENSO SST warming, highlighting the important role of the trade wind charging mechanism³⁶. By contrast, the SPMM warming directly generates SST warming on the equator,